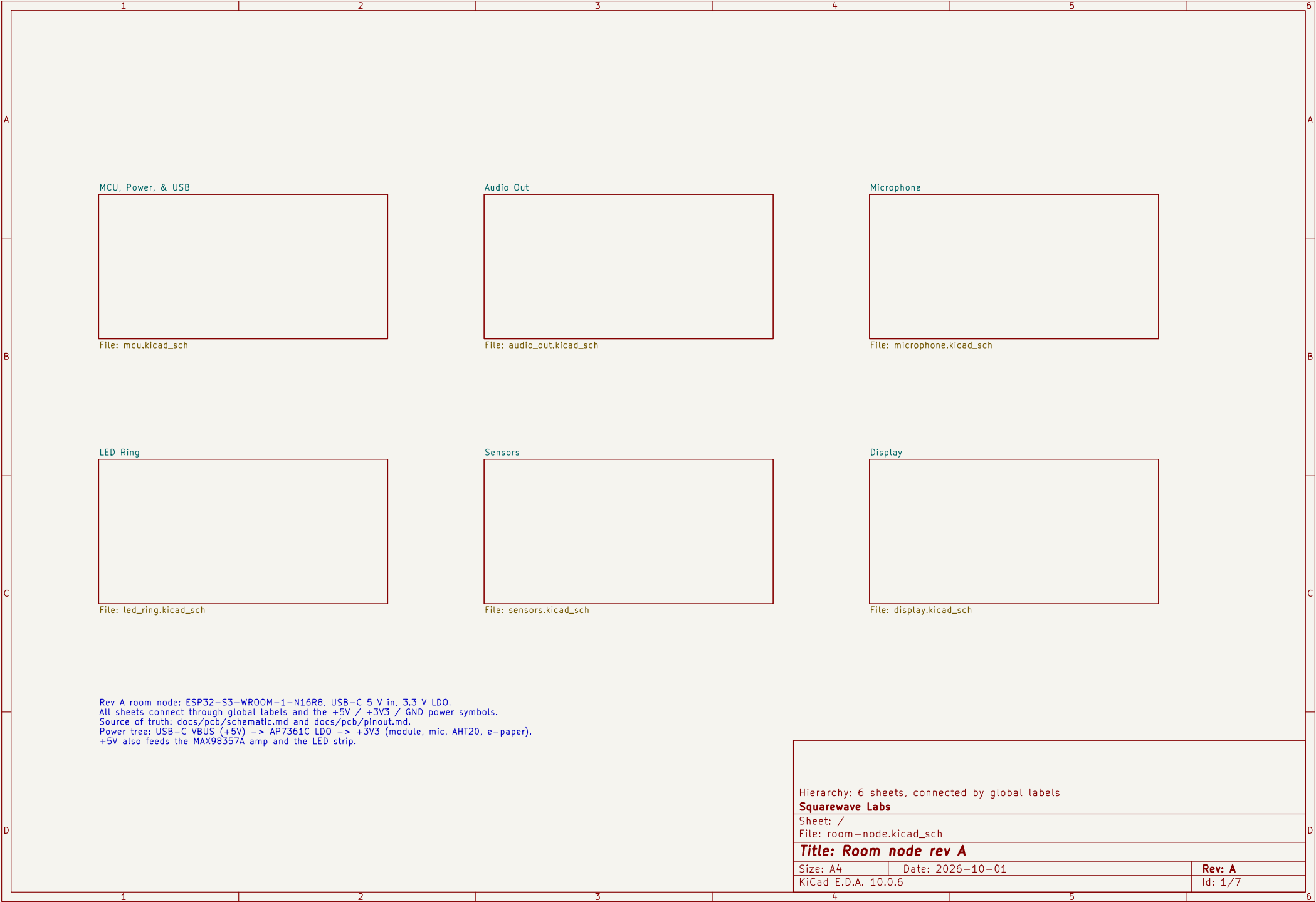




MCU, Power, & USB

File: mcu.kicad_sch

Audio Out

File: audio_out.kicad_sch

Microphone

File: microphone.kicad_sch

LED Ring

File: led_ring.kicad_sch

Sensors

File: sensors.kicad_sch

Display

File: display.kicad_sch

Rev A room node: ESP32-S3-WROOM-1-N16R8, USB-C 5 V in, 3.3 V LDO.
All sheets connect through global labels and the +5V / +3V3 / GND power symbols.
Source of truth: docs/pcb/schematic.md and docs/pcb/pinout.md.
Power tree: USB-C VBUS (+5V) -> AP7361C LDO -> +3V3 (module, mic, AHT20, e-paper).
+5V also feeds the MAX98357A amp and the LED strip.

Hierarchy: 6 sheets, connected by global labels

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Sheet: /

File: room-node.kicad_sch

Title: Room node rev A

Size: A4

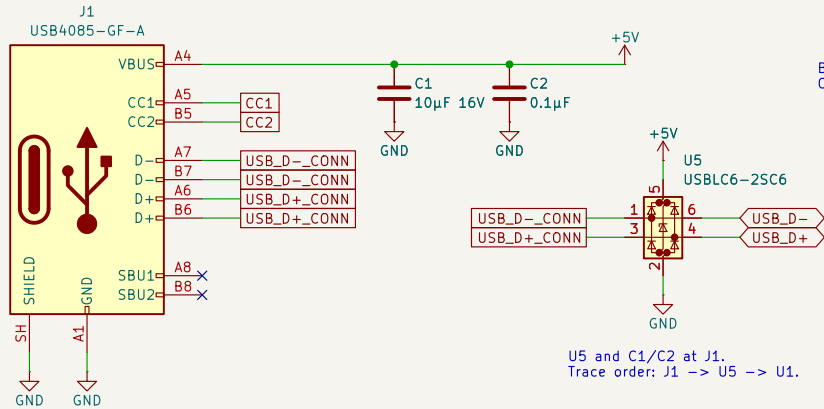
Date: 2026-10-01

Rev: A

KiCad E.D.A. 10.0.6

Id: 1/7

USB-C (5 V in) + ESD

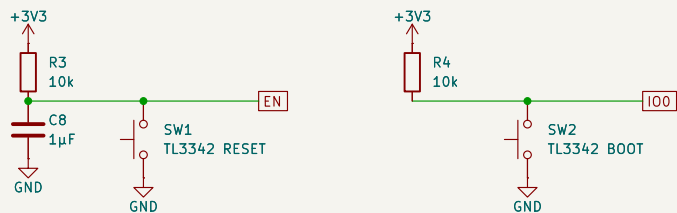


USB-C sink (Rd 5.1k) + CC sense



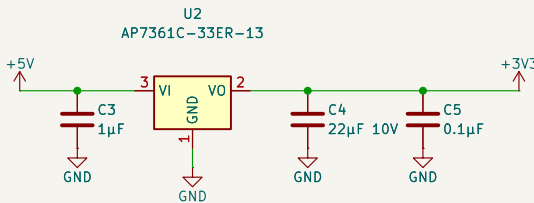
R6/R7 near J1. CC_SENSE ~0.2 V = 500 mA port.
0.46 V = 1.5 A, 0.84 V = 3 A (firmware limits LEDs).

Reset, boot, voice button



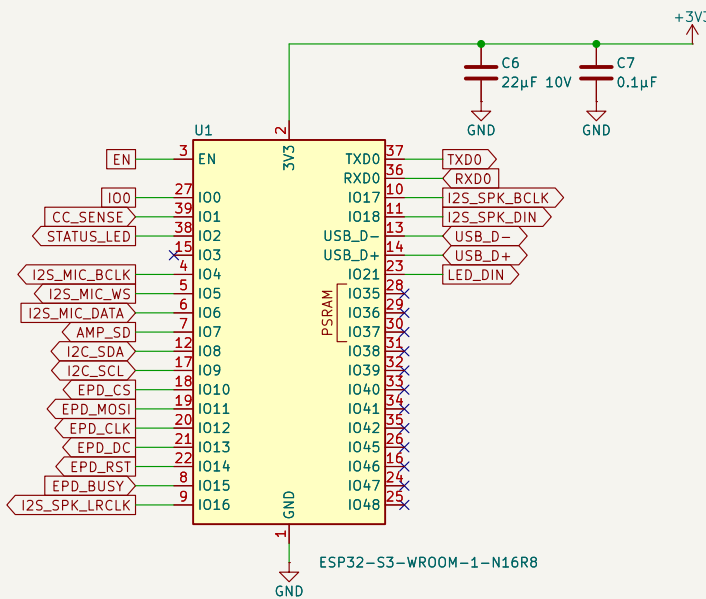
EN = U1 pin 3 (R3 10k + C8 1µF = power-on reset delay).
I00 = U1 pin 27, strapping pin; BOOT button only.
U1 pin 31 (GPIO38): not used (no voice button, wake word only).

3.3 V LDO



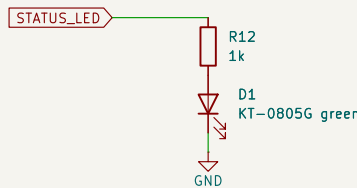
Buy the -33ER suffix only (SOT-223R: 1 GND, 2 OUT/tab, 3 IN).
Other suffixes have a different pinout. Copper pour on the tab.

ESP32-S3 module



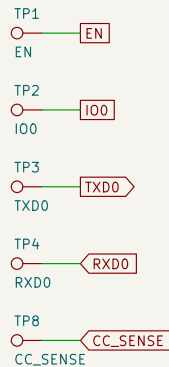
C6/C7 at U1 pin 2. Pins 1, 40, 41 (EP) to GND, via array under the pad.
Not connected: strapping GPIO3/45/46, PSRAM GPIO35-37,
spare GPIO39-42 (JTAG), 47, 48, GPIO2 = status LED. See pinout.md.
Keep the antenna keep-out free of copper on all layers.

Status LED (GPIO2)



U1 pin 38 (GPIO2), active HIGH. Not a strapping pin.
~0.5-0.9 mA (Vf 2.4-2.8 V at 1 mA). Firmware can dim it with PWM.
Put D1 where it shows through the case.

Test points (bring-up)



TP3 TXD0 / TP4 RXD0 = UART0 (U1 pins 37 / 36).
TP pads 1.5 mm, all on one edge.

Power flags (ERC only)



+5V comes from J1 VBUS (passive pins), so KiCad needs a PWR_FLAG to see it as driven. Same for GND.

Drawn by Iris (SQU-27). Chris reviews and tidies.
Source: docs/pcb/schematic.md Sheet 1, docs/pcb/pinout.md
Squarewave Labs
Sheet: /MCU, Power, & USB/
File: mcu.kicad_sch

Title: Room node rev A - MCU, power, USB

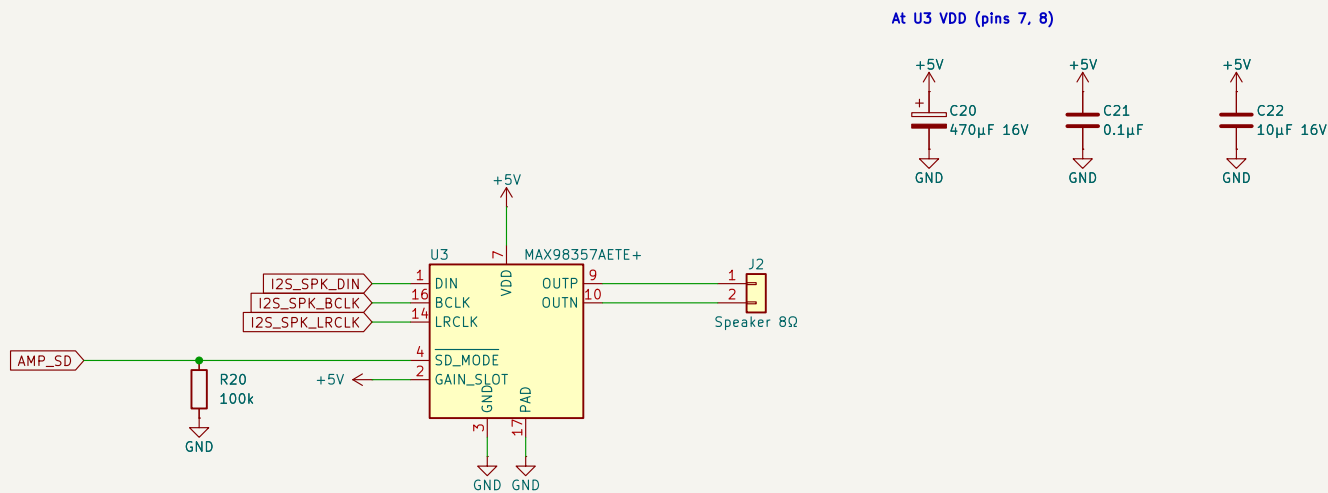
Size: A3 Date: 2026-10-02

KiCad E.D.A. 10.0.6

Rev: A

Id: 2/7

I2S class-D amplifier (MAX98357A) -> speaker



GAIN_SLOT to VDD = 6 dB gain (tested on the breadboard; 9 dB clips).
SD_MODE: R20 100k keeps the amp off during boot. GPIO HIGH = on, left channel.
EP (pin 17) is not connected inside the chip; wire it to GND, vias in the pad.
C20 + C21 + star ground removed the breadboard crackle. Keep all three, at the chip.
Speaker: Dayton CE32A-8 (8 Ω). Do not connect either speaker wire to GND.
NC pins 5, 6, 12, 13 are hidden on the stock symbol (no connect).

Drawn by Iris (SQU-27). Chris reviews and tidies.
Source: docs/pcb/schematic.md Sheet 2

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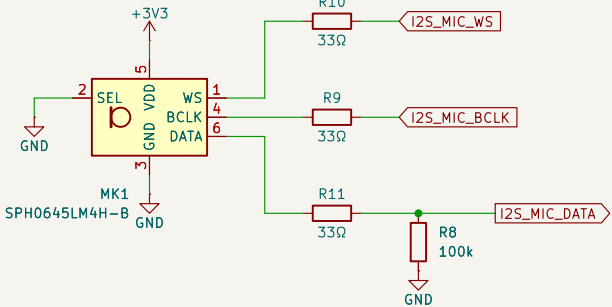
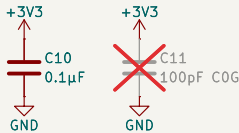
Sheet: /Audio Out/
File: audio_out.kicad_sch

Title: Room node rev A – Audio out

Size: A4	Date: 2026-10-01	Rev: A
KiCad E.D.A. 10.0.6		Id: 3/7

I2S MEMS microphone (SPH0645LM4H-B)

At MK1 VDD (pin 5)



SELECT (pin 2) to GND = left channel.
R9/R10 (33 Ω) at U1, R11 at MK1: optional damping; use 0 Ω if removed.
R8 100k on I2S_MIC_DATA to GND. C11 100 pF C0G is DNP (do not fit).
Use the stock symbol/footprint (Sensor_Audio). Footprint has the sound hole;
GND ring around it. Mic mute is software only (no mute switch).

Drawn by Iris (SQU-27). Chris reviews and tidies.
Source: docs/pcb/schematic.md Sheet 3

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Sheet: /Microphone/
File: microphone.kicad_sch

Title: Room node rev A – Microphone

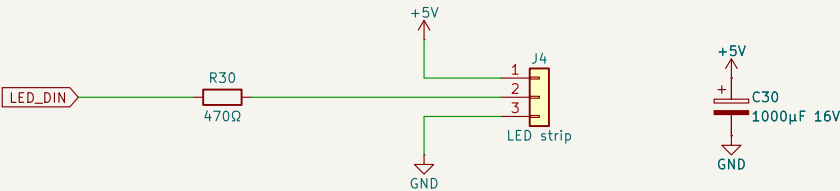
Size: A4 Date: 2026-10-01

KiCad E.D.A. 10.0.6

Rev: A

Id: 4/7

WS2812B LED strip connector



J4: 1 = +5V, 2 = DIN, 3 = GND. JST-XH side entry (S3B-XH-A), 3 A; full white is ~1.6 A.
R30 near U1. C30 across J4 pins 1 and 3, at J4.
Strip: 27 x WS2812B, off-board. No level shifter (works on the breadboard);
if the strip flickers on rev A, rev B adds a 74AHCT125.

Drawn by Iris (SQU-27). Chris reviews and tidies.
Source: docs/pcb/schematic.md Sheet 4

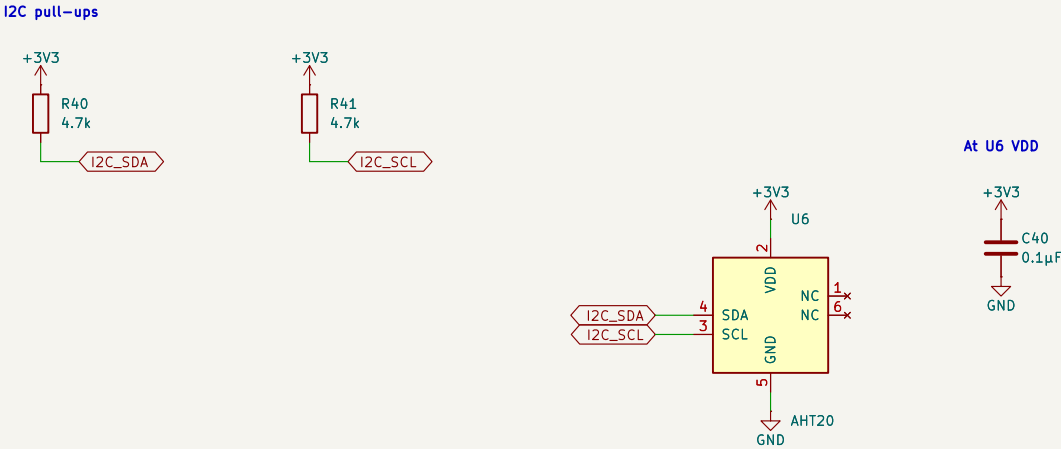
Squarewave Labs

Sheet: /LED Ring/
File: led_ring.kicad_sch

Title: Room node rev A – LED ring

Size: A4	Date: 2026-10-01	Rev: A
KiCad E.D.A. 10.0.6		Id: 5/7

Temperature / humidity (AHT20, I2C 0x38)



Pin 1 and 6 are NC on the AHT20 (no-connect pin type).
Project-library part re-made from LCSC C2757850 (6 pads, 1.0 mm pitch);
pinout per Aosong AHT20 datasheet v1.1 section 3.
Place U6 on a board edge, far from U1, U2, U3 and J4, with slots on three sides
(board heat reads as room temperature).

Drawn by Iris (SQU-27). Chris reviews and tidies.
Source: docs/pcb/schematic.md Sheet 5

Squarewave Labs

Sheet: /Sensors/
File: sensors.kicad_sch

Title: Room node rev A – Sensors

Size: A4

Date: 2026-10-01

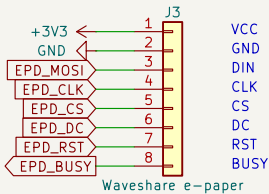
Rev: A

KiCad E.D.A. 10.0.6

Id: 6/7

Waveshare 2.9" e-paper module (with its driver board) via cable

At J3 pin 1



Check the pin order and count on your cable. Newer Waveshare modules have a 9th pin (PWR): if yours has it, use a 9-pin connector and tie PWR to +3V3.
Cable: needs a PH-to-PH 8-pin cable that maps pin 1 to pin 1 (beep it out before power-on). Option: 1x8 2.54 mm header (PinHeader_1x08_P2.54mm_Vertical) for the existing jumper wires. Same pin order either way.

Drawn by Iris (SQU-27). Chris reviews and tidies.
Source: docs/pcb/schematic.md Sheet 6

Squarewave Labs

Sheet: /Display/
File: display.kicad_sch

Title: Room node rev A – Display

Size: A4	Date: 2026-10-01	Rev: A
KiCad E.D.A. 10.0.6		Id: 7/7